DO NOW — RECALL

A*: heuristic reliability and route evaluation

Name: _____ Class: _____ Date: _____

1. Calculate f when g=4 and h=3.

2. What is the role of a predecessor table?

3. If S–A=2 and A–G=5, what is this route's total cost?

4. The true cheapest remaining cost from A is 4. Is h(A)=6 admissible?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

f=7.

2. Model answer

It lets us reconstruct the path by following predecessors backwards.

3. Model answer

7.

4. Model answer

No. It overestimates the true cost.

Vocabulary — know these terms

admissible — Heuristic never overestimates the true cheapest remaining cost.

consistent — Heuristic obeys h(u) ≤ edge(u,v)+h(v) for every edge.

suboptimal — A valid route whose cost is greater than the cheapest possible.

frontier — Discovered vertices awaiting expansion.

heuristic — An estimated remaining cost used to guide search.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Review the choice

O reaches G at g7,f7. G is selected before N, so the early-goal algorithm returns cost 7.

But M→N→O→G costs 2+2+2=6; the bad h(N)=7 overestimates N's true remaining cost 4.

Using an admissible estimate such as h(N)=3 allows the cheaper path to be found.

IMPROVE: Explain one correction to your own reasoning.

NEW LEARNING

Read, annotate and explain

Find the real remaining cost

From A, the cheapest route A→B→G costs 2+2=4. Therefore h(A)=3 is admissible, but h(A)=9 is not.

Safe estimates support optimal search

With h(A)=3, A is expanded before B, improving the route to B and finally G at cost 6.

Watch the misleading estimate

With h(A)=9, A has f=11; B has f=7 and leads to goal at cost 8.

Why the error matters

A* that stops on the first goal removed from the open set can return this 8-cost path before exploring A, even though a 6-cost route exists.

Check the guarantee

Admissibility means no overestimate. Consistency adds the edge test h(u) ≤ weight(u,v) + h(v). Without reopening settled nodes, A* requires consistency for its shortest-path guarantee. Explain this as well as the route cost.

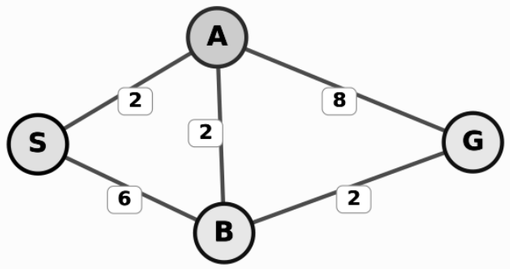
Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

SAFE h • FROM S



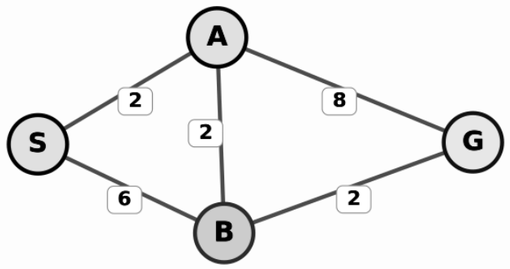
Node	g	h	f
A	2	3	5
B	6	1	7
G	∞	0	∞
S	0	5	5

A: f=5, B: f=7. Choose A.

SAFE h • FROM S; A: f=5, B: f=7. Choose A.

New Learning

SAFE h • VIA A



Node	g	h	f
A	2	3	5
B	4	1	5
G	10	0	10
S	0	5	5

B improves to g4, f5.

SAFE h • VIA A; B improves to g4, f5.

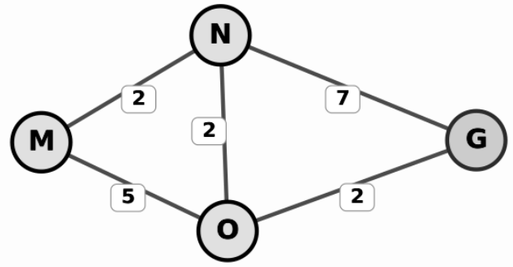
We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Can a bad estimate make A* return a longer route?

After M, O has f5+1=6; N has f2+7=9, so O is selected.

TRUE BEST PATH



Node	g	h	f
G	6	0	6
M	0	4	4
N	2	3	5
O	4	1	5

M→N→O→G costs 6

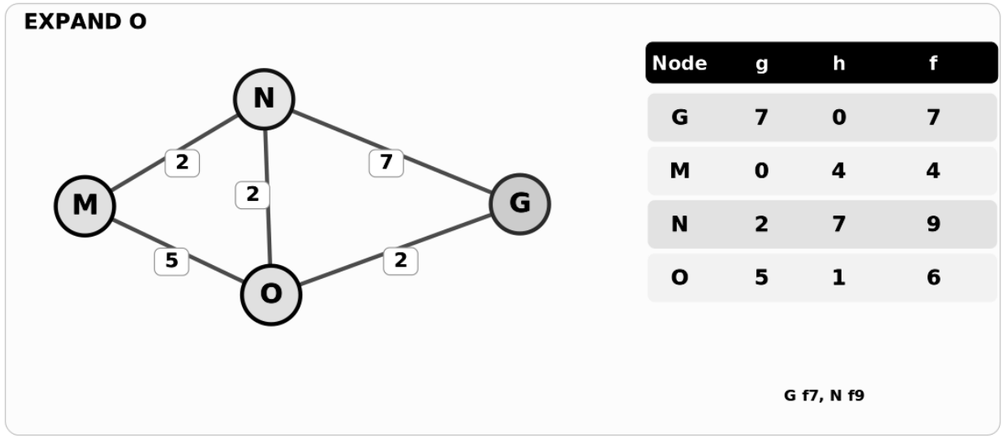
TRUE BEST PATH; M→N→O→G costs 6

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Can a bad estimate make A* return a longer route?

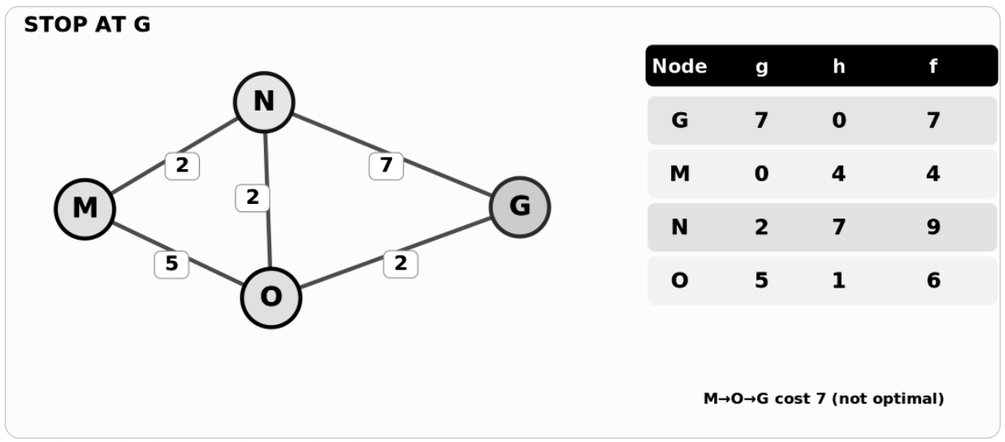
After M, O has f5+1=6; N has f2+7=9, so O is selected.



EXPAND O; G f7, N f9

Can a bad estimate make A* return a longer route?

After M, O has f5+1=6; N has f2+7=9, so O is selected.

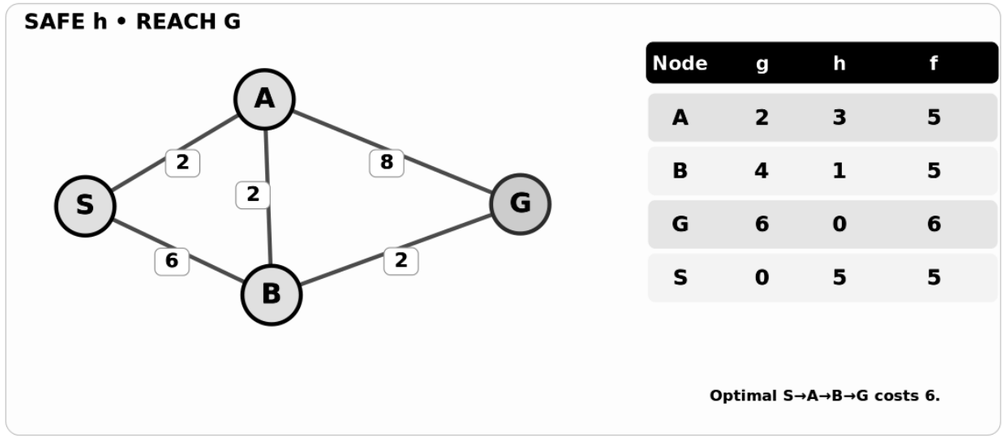


STOP AT G; M→O→G cost 7 (not optimal)

New Learning · original lesson diagrams

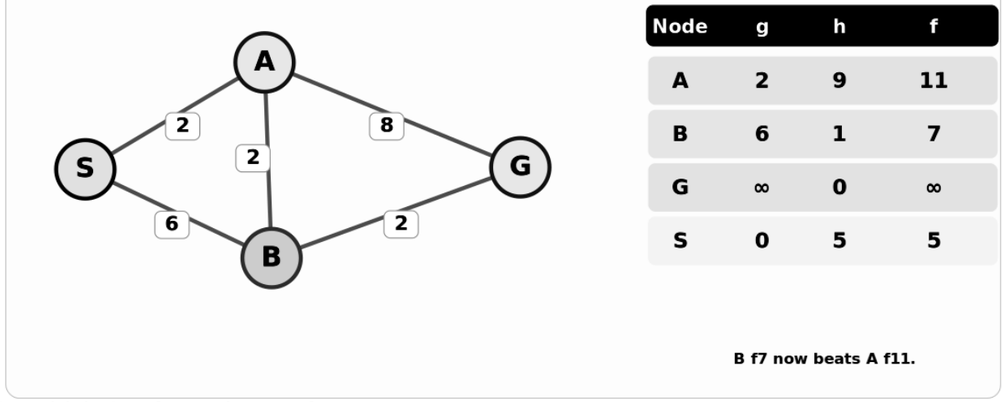
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



SAFE h • REACH G; Optimal S→A→B→G costs 6.

New Learning

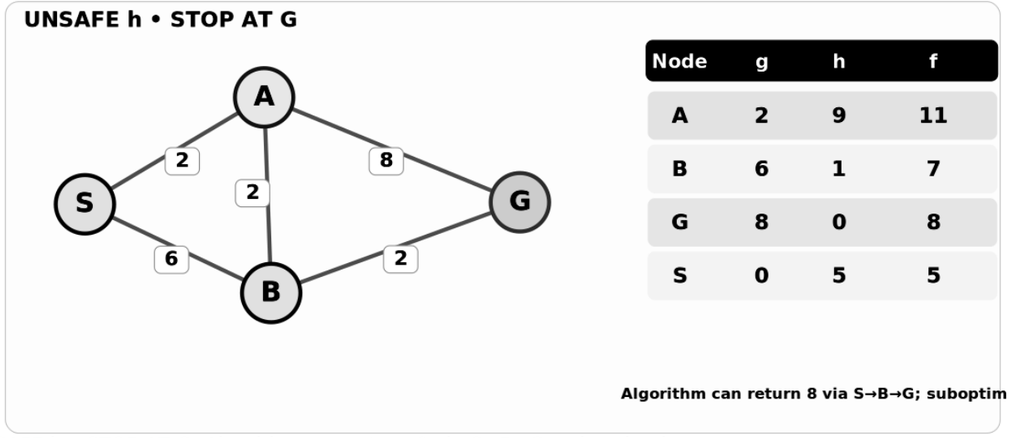


UNSAFE h(A)=9; B f7 now beats A f11.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



UNSAFE h • STOP AT G; Algorithm can return 8 via S→B→G; suboptimal.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

If a heuristic overestimates the true remaining cost, A* may lose its shortest-route guarantee.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare Dijkstra's algorithm and A* in their selection rule, heuristic use and guarantee of shortest paths.

I DO — WORKED EXAMPLE

Follow the worked method

Check admissibility and trace U→G with safe estimates.

Predict the choice

V has g1,h3,f4; W has g5,h1,f6. Expand V.

Review the choice

V improves W to g3 and f4; expand W.

Review the choice

W improves the goal G to g5,h0,f5.

Review the choice

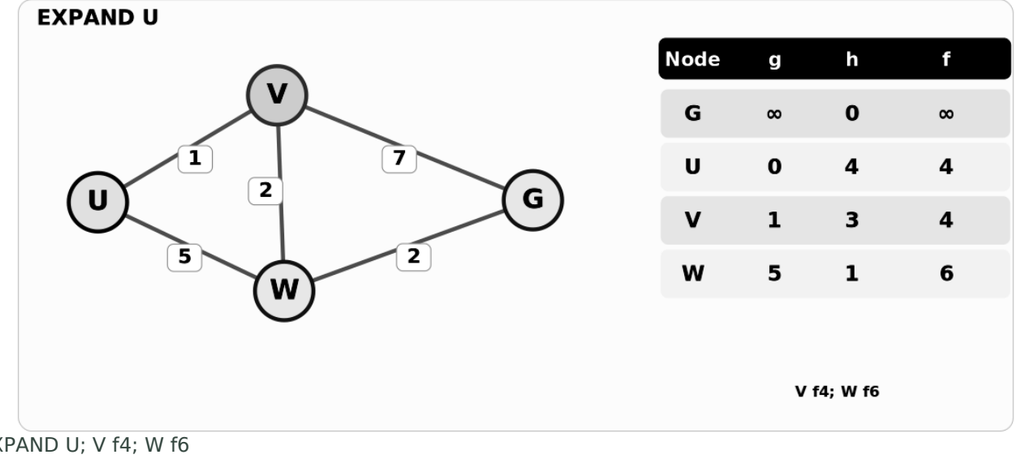
The path is U→V→W→G with total cost 5; the heuristics do not exceed the actual remaining costs.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Check admissibility and trace U→G with safe estimates.

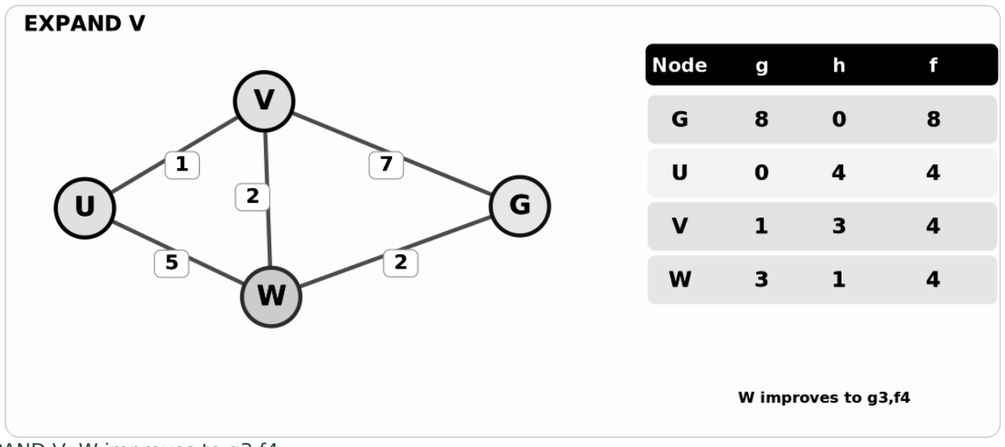
V has g1,h3,f4; W has g5,h1,f6. Expand V.



EXPAND U; V f4; W f6

Check admissibility and trace U→G with safe estimates.

V has g1,h3,f4; W has g5,h1,f6. Expand V.



EXPAND V; W improves to g3,f4

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

The true remaining distance from node A to goal G is 6. Decide whether h(A)=4 and h(A)=8 are admissible, with reasons.

Question 2

Edges S–A=2, S–B=6, A–B=2, B–G=2, A–G=8. Calculate the best S→G route and explain why h(A)=9 can mislead A*.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict the choice After M, O has f5+1=6; N has f2+7=9, so O is selected.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

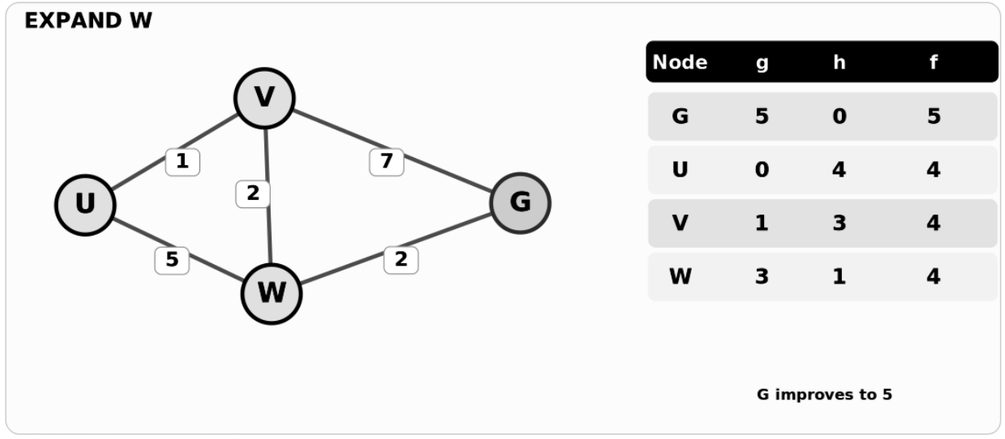
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Check admissibility and trace U→G with safe estimates.

V has g1,h3,f4; W has g5,h1,f6. Expand V.

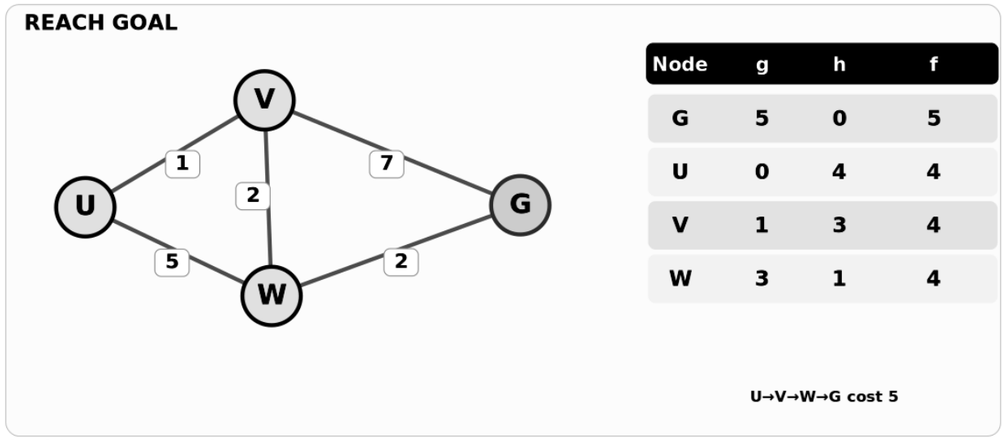


Node	g	h	f
G	5	0	5
U	0	4	4
V	1	3	4
W	3	1	4

EXPAND W; G improves to 5

Check admissibility and trace U→G with safe estimates.

V has g1,h3,f4; W has g5,h1,f6. Expand V.



Node	g	h	f
G	5	0	5
U	0	4	4
V	1	3	4
W	3	1	4

REACH GOAL; U→V→W→G cost 5

WE DO — GUIDED PRACTICE

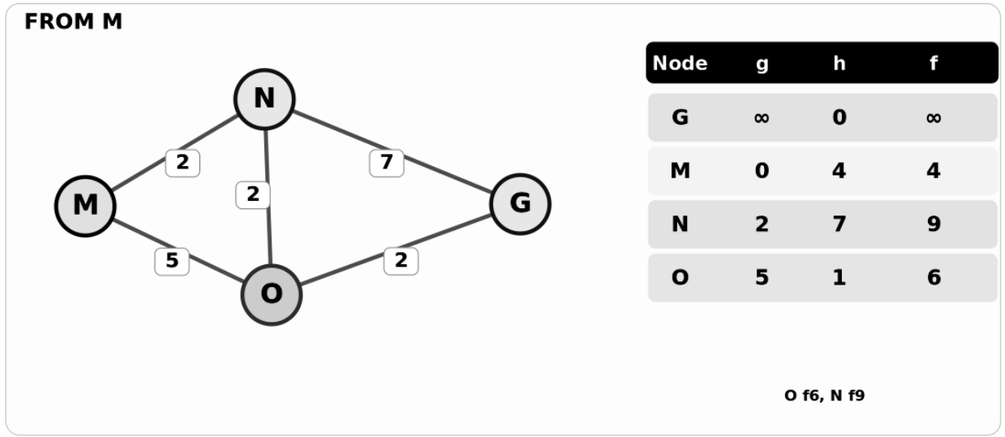
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Can a bad estimate make A* return a longer route?

After M, O has f5+1=6; N has f2+7=9, so O is selected.



FROM M; O f6, N f9